

workflows: does repeated self-application converge, how quickly does the residual edit magnitude decay, and what form of equilibrium emerges? We refer to this behavior as *textual relaxation*: a process in which repeated model revision drives text toward a model-preferred soft fixed-point region. Despite the growing adoption of recursive refinement in LLM systems, the relaxation dynamics of repeated revision remain largely unexplored.

Our perspective. In contrast to prior work, we treat recursive self-refinement itself as the primary object of study. Rather than asking only whether refinement improves final output quality, we investigate how refinement trajectories evolve under repeated application of the same refinement operator. We introduce exact and approximate fixed-point notions for text trajectories and provide empirical evidence that recursive self-refinement rapidly approaches a soft fixed-point region, with edit magnitudes following an exponential relaxation pattern. More broadly, our work shifts the focus from task-level improvement to the dynamical properties of recursive language-model behavior, including convergence, saturation, residual fluctuations, and practical stopping criteria.

3 Method

We study recursive self-refinement as a discrete dynamical process over text. Given an initial abstract V_0 , we repeatedly apply the same refinement operator R_θ , implemented by an LLM under a fixed prompt and decoding configuration:

$$V_{t+1} = R_\theta(V_t), \quad t = 0, \dots, T-1.$$

This produces a refinement trajectory $V_0, V_1, \dots, V_T$. In our experiments, $T = 10$, and the prompt instructs the model to improve clarity, coherence, conciseness, and scientific writing quality while preserving all technical content, claims, methods, results, and numerical values. This setup lets us study the behavior induced by repeated self-application of the same revision operator.

Transition magnitude. To measure how much each refinement step changes the text, we compute the normalized Levenshtein edit distance between consecutive versions:

$$\Delta_t = \frac{\text{EditDistance}(V_t, V_{t+1})}{\max(|V_t|, |V_{t+1}|, 1)}.$$

We interpret Δ_t as the transition magnitude or residual edit pressure at iteration t . Smaller values indicate that consecutive versions are closer, while sustained small values indicate saturation of the refinement process. We also track word counts and relative word-count change to distinguish minor surface edits from substantive compression or expansion.

Exact and approximate fixed points. An exact fixed point occurs when the refinement operator leaves the text unchanged:

$$V_{t+1} = V_t.$$

However, exact equality is often too strict for natural language. Even after a text is essentially stable, a model may still alter punctuation, formatting, or local phrasing. We therefore define approximate convergence using both edit distance and length stability. A transition is approximately converged if

$$\Delta_t < \epsilon_{\text{edit}}$$

and

$$\frac{|\text{wc}(V_{t+1}) - \text{wc}(V_t)|}{\max(\text{wc}(V_t), 1)} < \epsilon_{\text{wc}}.$$

We use $\epsilon_{\text{edit}} = 0.02$ and $\epsilon_{\text{wc}} = 0.03$. A trajectory is said to reach a soft fixed point at iteration t if this condition holds for two consecutive transitions starting at t . This criterion captures convergence to a stable textual region rather than exact string equality, and provides a practical stopping rule for recursive refinement workflows.

Textual relaxation model. To quantify how edit magnitudes decay over repeated refinement, we fit an exponential relaxation model to the average transition magnitude:

$$\Delta_t \approx Ae^{-kt} + c.$$

Here, A captures the initial edit pressure, k is the empirical convergence rate, and c represents a residual fluctuation floor near the soft fixed-point region. We use this model as a compact description of textual relaxation: a rapidly saturating process in which repeated revision approaches a model-preferred stable region. The model is not intended to imply that LLM refinement follows a literal physical law or performs gradient descent on an explicit objective.

External judge evaluation. Distance metrics capture textual stability but not whether refinement improves quality. We therefore perform an external LLM-as-a-judge evaluation on a subset of trajectories. A Gemini judge compares V_0 , V_1 , and V_T for each sampled abstract and rates clarity, conciseness, technical fidelity, scientific style, unnecessary change, and meaning preservation. This evaluation tests whether convergence corresponds to improved scientific writing rather than mere textual stagnation.

4 Experiments

Datasets. We evaluate recursive self-refinement on scientific abstracts from ICML proceedings. Our main dataset contains 50 abstracts from ICML 2025, collected from the PMLR proceedings. To test whether the observed dynamics are specific to recent LLM-era writing, we also collect a smaller cross-year comparison set of 15 abstracts from ICML 2020. Each example consists of a paper title and abstract. The title is provided to the model at every refinement step, while only the abstract is recursively revised.

Refinement model and prompt. Our main refinement model is GPT-5.5. For each abstract, we generate a trajectory of $T = 10$ refinement iterations. The model is prompted to act as an expert scientific editor preparing abstracts for a top-tier machine learning conference. The prompt asks the model to improve clarity, coherence, conciseness, and scientific writing quality while preserving all technical content, claims, methods, results, and numerical values. The model is instructed to return only the revised abstract.

Decoding settings. For ICML 2025, we run two decoding conditions: the API default-temperature setting and deterministic decoding with temperature 0. In both settings, we disable explicit reasoning by setting reasoning effort to none and cap the output length at 700 tokens. For the ICML 2020 comparison, we use the deterministic temperature 0 setting. All experiments use the same prompt and refinement horizon.

Evaluation metrics. For each transition $V_t \rightarrow V_{t+1}$, we compute normalized Levenshtein edit distance, word count, relative word-count change, and whether the text changed exactly. For each trajectory, we report the first exact fixed point, the first approximate fixed point, the mean and maximum normalized edit distance, and the final-step normalized edit distance. Approximate convergence uses $\epsilon_{\text{edit}} = 0.02$, $\epsilon_{\text{wc}} = 0.03$, and requires two consecutive approximately converged transitions.

External judge evaluation. To assess whether textual relaxation corresponds to quality improvement, we perform an external LLM-as-a-judge evaluation on 10 sampled ICML 2025 trajectories from the temperature 0 condition. The judge is Gemini 3.5 Flash. For each sampled abstract, the judge compares V_0 , V_1 , and V_{10} , scoring clarity, conciseness, technical fidelity, scientific style, unnecessary change, and meaning preservation on a 1–5 scale, and selecting the best version.

5 Results

Figure 2 visualizes the empirical convergence trajectory by projecting each refinement iteration according to its average closeness to the final version, $1 - d(V_t, V_{10})$. The trajectory rapidly enters a soft fixed-point region after the first few iterations, with only small residual movement from V_8 to V_{10} .

Recursive refinement converges rapidly. Table 1 summarizes the main convergence results. On 50 ICML 2025 abstracts, GPT-5.5 reaches approximate convergence for all trajectories under both

Condition	n	Exact rate	Approx. rate	Mean t_{exact}^*	Mean t_{approx}^*	Final Δ_t	Zero-final rate
ICML 2025, default	50	0.94	1.00	4.66	2.42	0.0055	0.56
ICML 2025, temp. 0	50	1.00	1.00	2.60	1.38	0.0003	0.92
ICML 2020, temp. 0	15	1.00	1.00	3.00	1.67	0.0010	0.80

Table 1: Convergence summary for recursive refinement. Exact rate is the fraction of trajectories reaching exact textual fixed points. Approximate rate uses $\epsilon_{\text{edit}} = 0.02$, $\epsilon_{\text{wc}} = 0.03$, and two consecutive converged transitions. Final Δ_t is the mean normalized edit distance at $V_9 \rightarrow V_{10}$. Zero-final rate is the fraction of trajectories with no final-step textual change.

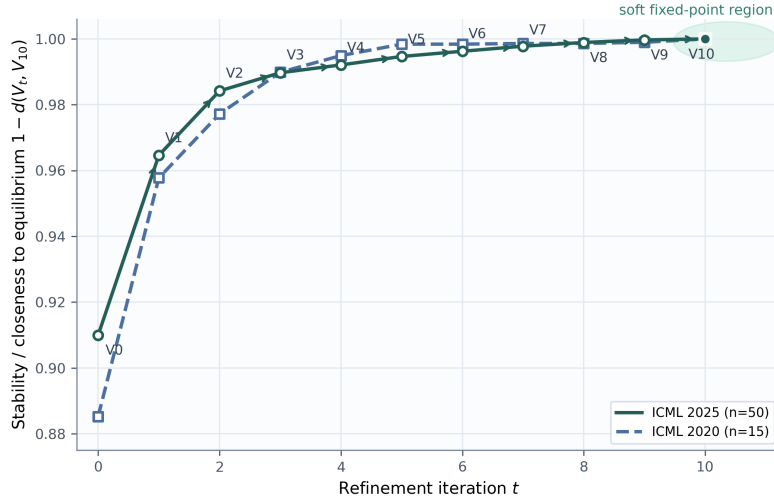


Figure 2: Both ICML 2020 and ICML 2025 abstracts exhibit qualitatively similar convergence trajectories. Most movement occurs during the first few refinement iterations, after which trajectories rapidly approach a stable soft fixed-point region with only minor residual changes.

default-temperature and temperature 0 decoding. Exact fixed points are also common: 94% of default-temperature trajectories and 100% of temperature 0 trajectories reach exact textual fixed points within 10 iterations. Approximate convergence occurs earlier than exact convergence, with mean approximate convergence times of 2.42 and 1.38 transitions for default-temperature and temperature 0, respectively.

Temperature controls residual fluctuations. Figure 3 shows that most editing occurs in the first few refinement steps. Under temperature 0, the mean normalized edit distance drops from 0.0592 at $V_0 \rightarrow V_1$ to a final-step value of 0.0003 at $V_9 \rightarrow V_{10}$. Default-temperature decoding exhibits the same qualitative decay but with larger residual fluctuations. Thus, deterministic decoding more often collapses to exact textual fixed points, while default-temperature decoding remains in a noisier soft fixed-point region.

Edit magnitudes follow a relaxation-like curve. The average transition magnitude is well fit by an exponential relaxation model,

$$\Delta_t \approx Ae^{-kt} + c.$$

For the ICML 2025 default-temperature condition, the fitted curve achieves $R^2 = 0.990$, with $A = 0.0585$, $k = 0.923$, and residual floor $c = 0.0070$. This supports the view that recursive refinement behaves like a rapid relaxation process toward a soft textual equilibrium rather than an open-ended optimization procedure. This saturation suggests a practical stopping criterion: recursive refinement can be terminated once Δ_t remains below a small threshold for consecutive iterations.

External judging supports quality improvement. To verify that convergence is not merely textual stagnation, we evaluate 10 sampled trajectories using Gemini 3.5 Flash as an external judge. Gemini